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DAVID CHENG

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SUMMARY

Versatile software engineer with 3.5 years of work experience developing real-time C++ software for agricultural machinery. Experienced with backend design patterns applied to GUIs (MVVM, publish–subscribe, observer, hexagonal architecture) and delivering end-to-end features across requirements gathering, coding, documentation, and hardware validation in an Agile environment.

SKILLS

Languages	C++/C, Java, Python
Software	Git, GitLab, CMake, Linux, Qt Framework, QML, GoogleTest, GDB, MATLAB
Tools	Jira, Polarion, Confluence
Embedded	QEMU, Arduino, ESP32, ESP8266, Raspberry Pi Pico

WORK EXPERIENCE

CNH Industrial Oak Brook, IL

Embedded Software Engineer II – *Guidance Team* July 2024 - Present

- Worked on the precision farming backend component in C++, processing operator inputs and real-time guidance data for UI visualization in a Qt application on an embedded Linux using Yocto.
- Developed QML UIs by binding backend C++ data to the ViewModel with Qt signals and slots, enabling operators to use guidance features in sync with streamed data using the MVVM pattern.
- Implemented real-time guidance line tracking by integrating swath generation and display APIs using a dirty-flag with observer pattern, updating only on changed data to improve efficiency.
- Improved operator setup and load times by 20% by offloading persistence operations to a separate thread, ensuring critical events are handled and displayed to the UI without delay.
- Enabled USB-based import of agricultural swath data by parsing and converting ISOXML files to a standardized format, allowing operators to reuse agricultural data across machines.
- Delivered end-to-end features in a SAFe-based Agile environment, from scoping requirements with system engineers and TPM, through implementation in C++, to validation on QEMU and target hardware using GDB.

Embedded Software Engineer I – *Automatic End-of-Row Turn (AEORT) Project* July 2022 - July 2024

- Led the development of an automated field turn planner using a Dubins-Clothoid turning algorithm to generate shortest drivable turns between agricultural swaths.
- Designed a dynamic turn planning solution for collision turns by dividing them into sub-turns for re-planning, resulting in a formal IP submission.
- Increased test coverage for newly developed software by 30% by writing GoogleTest unit tests and mocking dependencies, improving algorithm reliability.
- Documented software at different levels using Doxygen, Markdown, Mermaid diagrams, improving clarity throughout the development process.

EDUCATION

New York Institute of Technology Sept 2017 - May 2022

M.S. Electrical & Computer Engineering GPA: 3.96/4.00

B.S. Electrical & Computer Engineering GPA: 3.89/4.00

Degree Honors: *summa cum laude*, Presidential Honor List (2018-2021)

GitHub Portfolio (Academic & Early-Career Projects)**:** https://daviecheng.github.io/